

Lecture 6 The Bayes' Theorem

BIO210 Biostatistics

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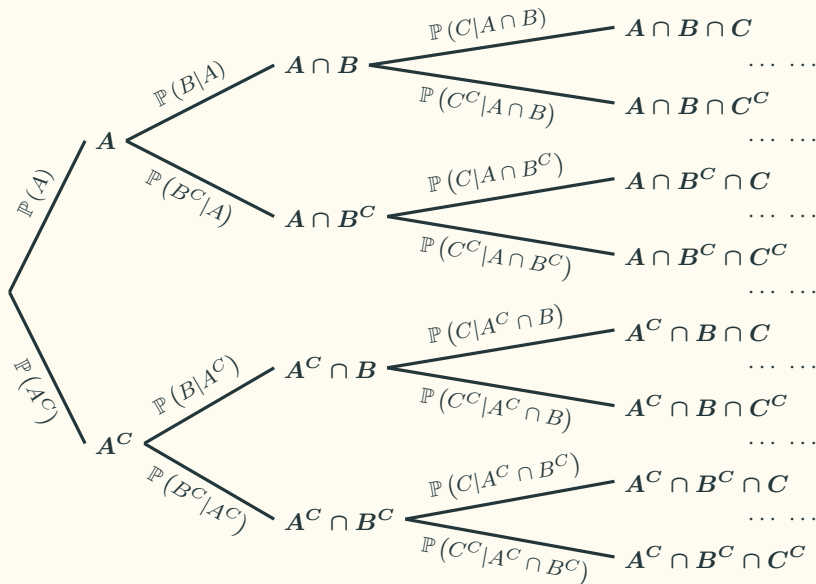


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Three basic components in conditional probability:

1. $\mathbb{P}(A \cap B)$
2. $\mathbb{P}(B)$
3. $\mathbb{P}(A|B)$

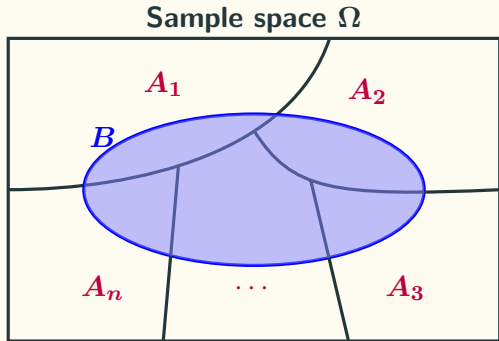
Generalisation of $\mathbb{P}(A \cap B \cap C \cap \dots)$



The multiplication rule:

$$\begin{aligned}
 &\mathbb{P}\left(\bigcap_{i=1}^n A_i\right) = \\
 &\mathbb{P}(A_1) \cdot \\
 &\mathbb{P}(A_2|A_1) \cdot \\
 &\mathbb{P}(A_3|A_1 \cap A_2) \cdot \\
 &\mathbb{P}(A_4|A_1 \cap A_2 \cap A_3) \cdot \\
 &\dots \\
 &\mathbb{P}\left(A_n \mid \bigcap_{i=1}^{n-1} A_i\right)
 \end{aligned}$$

Generalisation of $\mathbb{P}(B)$

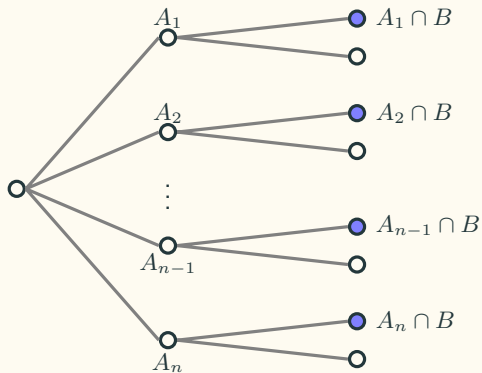


$$\begin{aligned}\mathbb{P}(B) &= \mathbb{P}[(A_1 \cap B) \cup (A_2 \cap B) \cup \dots \cup (A_n \cap B)] \\ &= \mathbb{P}(A_1 \cap B) + \mathbb{P}(A_2 \cap B) + \dots + \mathbb{P}(A_n \cap B) \\ &= \mathbb{P}(A_1) \mathbb{P}(B|A_1) + \\ &\quad \mathbb{P}(A_2) \mathbb{P}(B|A_2) + \\ &\quad \vdots \\ &\quad \mathbb{P}(A_n) \mathbb{P}(B|A_n)\end{aligned}$$

The total probability theorem:

$$\mathbb{P}(B) = \sum_{i=1}^n \mathbb{P}(B \cap A_i) = \sum_{i=1}^n \mathbb{P}(A_i) \mathbb{P}(B|A_i)$$

Generalisation of $\mathbb{P}(B)$

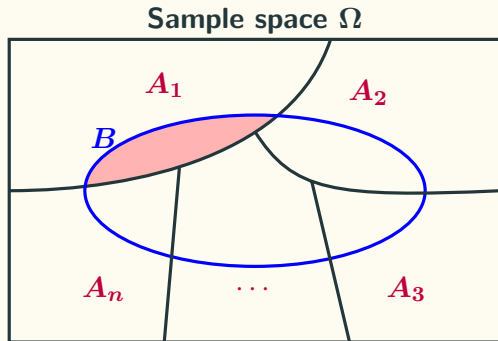


$$\begin{aligned}\mathbb{P}(B) &= \mathbb{P}[(A_1 \cap B) \cup (A_2 \cap B) \cup \dots \cup (A_n \cap B)] \\ &= \mathbb{P}(A_1 \cap B) + \mathbb{P}(A_2 \cap B) + \dots + \mathbb{P}(A_n \cap B) \\ &= \mathbb{P}(A_1) \mathbb{P}(B|A_1) + \\ &\quad \mathbb{P}(A_2) \mathbb{P}(B|A_2) + \\ &\quad \vdots \\ &\quad \mathbb{P}(A_n) \mathbb{P}(B|A_n)\end{aligned}$$

The total probability theorem:

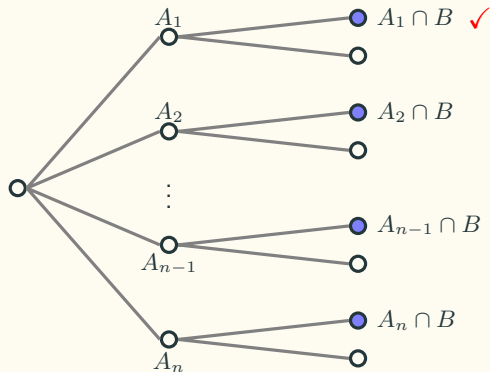
$$\mathbb{P}(B) = \sum_{i=1}^n \mathbb{P}(B \cap A_i) = \sum_{i=1}^n \mathbb{P}(A_i) \mathbb{P}(B|A_i)$$

Generalisation of $\mathbb{P}(A|B)$



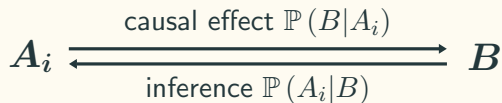
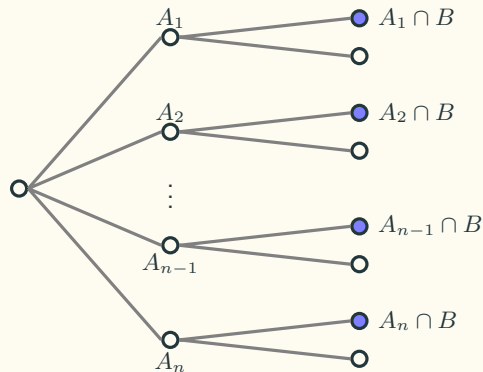
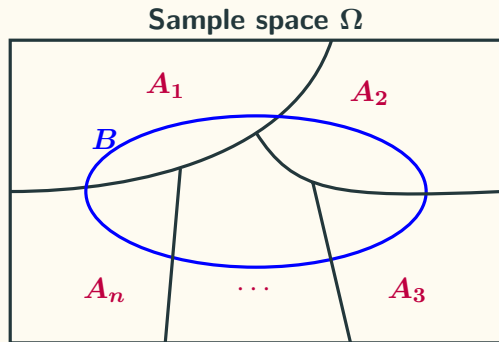
$$\begin{aligned}\mathbb{P}(A_i|B) &= \frac{\mathbb{P}(A_i \cap B)}{\mathbb{P}(B)} \\ &= \frac{\mathbb{P}(A_i) \mathbb{P}(B|A_i)}{\mathbb{P}(B)} \\ &= \frac{\mathbb{P}(A_i) \mathbb{P}(B|A_i)}{\sum_{i=1}^n \mathbb{P}(A_i) \mathbb{P}(B|A_i)}\end{aligned}$$

Generalisation of $\mathbb{P}(A|B)$



$$\begin{aligned}\mathbb{P}(A_i|B) &= \frac{\mathbb{P}(A_i \cap B)}{\mathbb{P}(B)} \\ &= \frac{\mathbb{P}(A_i) \mathbb{P}(B|A_i)}{\mathbb{P}(B)} \\ &= \frac{\mathbb{P}(A_i) \mathbb{P}(B|A_i)}{\sum_{i=1}^n \mathbb{P}(A_i) \mathbb{P}(B|A_i)}\end{aligned}$$

The Bayes Theorem



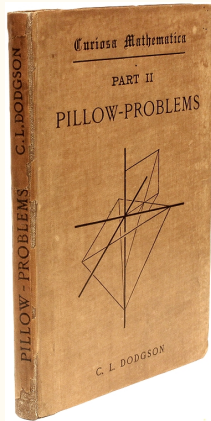
The Bayes' Theorem

$$\mathbb{P}(A_i|B) = \frac{\mathbb{P}(A_i \cap B)}{\mathbb{P}(B)} = \frac{\mathbb{P}(A_i) \mathbb{P}(B|A_i)}{\sum_{i=1}^n \mathbb{P}(A_i) \mathbb{P}(B|A_i)}$$

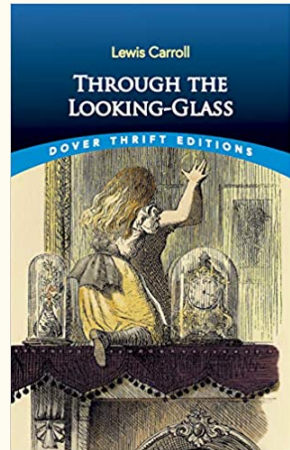
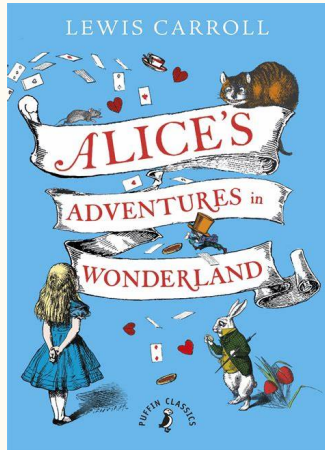
$\mathbb{P}(A_i)$: **prior probability**

$\mathbb{P}(A_i|B)$: **posterior probability**

Lewis Carroll's Pillow Problems



by Charles Dodgson



Question #5 (8th Sep 1887):

2

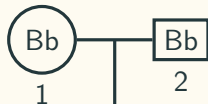
PILLOW-PROBLEMS.

5. (19, 31)

A bag contains one counter, known to be either white or black. A white counter is put in, the bag shaken, and a counter drawn out, which proves to be white. What is now the chance of drawing a white counter? [8/9/87]

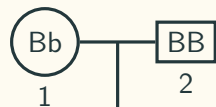
Pedigree Analysis

Generation I



Generation II

Generation I



Generation II



Generation III

